SNAPDRAGON SENSE

AI-Powered Laptop Troubleshooting Assistant

Project Description | Prototype Submission

Project summary

Snapdragon Sense is a prototype concept for an AI assistant that helps laptop users understand common symptoms—such as unusual battery drain, slow performance, app crashes, freezes, and wireless problems—and decide what to check next. It converts user-provided observations into plain-language troubleshooting guidance.

Problem

When a laptop behaves unexpectedly, users often encounter technical error messages and generic online advice. They may not know which evidence to collect, which checks are safe, or which actions could make the problem worse. The prototype focuses on making the first troubleshooting steps clearer and more structured.

Proposed solution

A lightweight interface collects the user's symptom and optional details. A guided analysis layer produces cautious, low-risk checks. An optional local AI mode uses an open-source model served by Ollama on the same computer to explain the assessment in clear language. The prototype does not silently monitor the PC or change system settings.

Intended Snapdragon fit

The solution is intended for future optimization on Snapdragon-powered HP PCs. A local inference architecture can reduce dependence on cloud services and keep diagnostic prompts on the device. Actual Snapdragon model compatibility, runtime selection, latency, and power consumption must be tested before claiming performance or NPU acceleration.

Current prototype capabilities

- Streamlit interface with guided paths for common symptom categories.
- User-entered diagnostics and an explicitly labelled sample demo.
- Optional local model integration through Ollama.
- Fallback guidance when a local model is not configured.
- Clear limitations and no automatic system modifications.

AI use and user journey

The local model is used to explain the user-provided symptom and generate structured troubleshooting guidance. The user selects a symptom, adds optional details, receives a cautious assessment, reviews suggested low-risk checks, and decides whether to follow them or consult official support.

Safety, privacy, and limitations

The prototype uses only user-provided text and selected values. It does not collect live system telemetry, access personal files, or change system settings. In local AI mode, prompt text is sent to the local Ollama service. Recommendations are hypotheses, not confirmed diagnoses.

Validation plan

Test representative scenarios; add consent-based access to supported Windows diagnostics; select and document a model from Qualcomm AI Hub or another open-source platform; test on actual Snapdragon-powered HP hardware; and record repeatable latency, answer quality, memory, and power measurements.

Current status

Proof of concept. Packaged for local demonstration on Windows. Snapdragon hardware testing and deployment optimization are not yet completed.